INTRODUCTION TO CLOUD COMPUTING

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Why Learn Cloud Computing?

- High Industry Demand
- Cost Efficiency
- Scalability & Flexibility
- Faster Development & Deployment
- Global Access & Collaboration
- Reliability & Disaster Recovery
- Supports Modern Technologies
- Career Growth & Future Proof Skills

What is cloud computing ?

Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing.

Key Characteristics

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service (metered billing)
- Multi-Tenancy
- Virtualization
- Resiliency and Availability

Cloud Deployment Models

- Public Cloud
- Private Cloud
- Hybrid Cloud
- Multi-Cloud

Public Cloud

A **Public Cloud** is a cloud computing model where **services are provided over the internet** by a third-party cloud provider and **shared among multiple customers**.

Infrastructure (servers, storage, networking) is **owned and managed by the cloud provider**.

Key Characteristics

- Shared infrastructure (multi-tenant)
- Pay-as-you-go pricing
- High scalability and elasticity
- Managed and maintained by provider
- Accessible from anywhere via internet

Public Cloud Services

- **IaaS:** Virtual machines, storage, networks
- **PaaS:** Application platforms, databases
- **SaaS:** Email, CRM, office tools

Use Cases of Public Cloud

- Web & mobile application hosting
- Startup platforms
- Development & testing environments
- Big data analytics
- AI/ML model training
- E-learning platforms

Advantages of Public Cloud

- Low initial cost (no hardware purchase)
- Easily scalable
- Fast deployment
- Global availability
- Provider handles maintenance & updates

Disadvantages of Public Cloud

- Less control over infrastructure
- Data security and privacy concerns
- Possible vendor lock-in
- Internet dependency

Private Cloud

A **Private Cloud** is a cloud computing environment **dedicated to a single organization**.
The infrastructure is **not shared** with other users and provides **greater control, security, and customization**.

It can be:

- Hosted **on-premise**, or
- Hosted by a **third-party provider** but used exclusively by one organization

Key Characteristics

- Single-tenant environment
- High security and privacy
- Full control over infrastructure
- Customizable configurations
- Limited but predictable scalability

Examples of Private Cloud Technologies

- OpenStack
- VMware Cloud
- Microsoft Azure Stack
- AWS Outposts

Private Cloud Services

- **IaaS:** Dedicated virtual machines, storage, networking
- **PaaS:** Private application platforms
- **Internal SaaS:** Company-specific software

Use Cases of Private Cloud

- Banking and financial systems
- Healthcare systems (patient records)
- Government applications
- Large enterprises with sensitive data
- Mission-critical applications

- **Advantages of Private Cloud**
- Strong data security and compliance
- Greater control and customization
- Better performance consistency
- Suitable for regulatory requirements (HIPAA, GDPR)

- **Disadvantages of Private Cloud**
- High setup and maintenance cost
- Requires skilled IT team
- Limited scalability compared to public cloud
- Slower deployment

Hybrid Cloud

- **What Is a Hybrid Cloud?**
- A **Hybrid Cloud** combines **two or more different environments**:
  - **Private Cloud / On-Premise infrastructure**
  - **Public Cloud**
- These environments are **integrated and work together**, allowing data and applications to move between them.

Key Characteristics

- Mix of public and private cloud
- Secure data stays on private cloud
- Public cloud used for scalability
- Workload portability
- Centralized management

Hybrid Cloud Architecture

- Sensitive data → **Private Cloud**
- High traffic or compute-intensive tasks → **Public Cloud**
- Secure connection between environments (VPN / Direct Connect)

Why Use Hybrid Cloud?

- Maintain control over sensitive data
- Scale applications when demand increases
- Modernize legacy systems gradually
- Meet regulatory and compliance requirements

Use Cases of Hybrid Cloud

- Banking & financial systems
- Healthcare applications
- Government services
- E-commerce during peak seasons
- Enterprise legacy system modernization

Advantages of Hybrid Cloud

- Best of both worlds (security + scalability)
- Cost optimization
- High flexibility
- Business continuity & disaster recovery
- Better compliance management

Challenges of Hybrid Cloud

- Complex architecture
- Integration and networking issues
- Security management across environments
- Requires skilled IT teams

Technologies Used in Hybrid Cloud

- Kubernetes
- VPN / Direct Connect
- Terraform
- Identity & Access Management (IAM)
- Monitoring & logging tools

- **Real-World Example**
- Banking System
  - Customer data stored in private cloud
  - Public cloud used for analytics & mobile apps
  - Secure integration between both
- **When to Use Hybrid Cloud?**
  - When data privacy is critical
  - When gradual cloud migration is needed
  - When regulatory compliance is required

Multi-Cloud

What Is Multi-Cloud?

- **Multi-Cloud** is a cloud strategy where an organization uses **services from multiple cloud providers** at the same time.
- Example:
  Using **AWS for compute**, **Azure for enterprise apps**, and **Google Cloud for analytics**.

Key Characteristics

- Uses more than one cloud provider
- Can include public and private clouds
- Avoids dependency on a single vendor
- Each cloud used for its strengths

Why Use Multi-Cloud?

- Avoid vendor lock-in
- Improve reliability & availability
- Optimize cost
- Use best-of-breed services
- Meet regional or legal requirements

Use Cases of Multi-Cloud

- Global applications with regional compliance
- Disaster recovery across providers
- Large enterprises with diverse workloads
- High-availability systems
- AI/ML workloads using specialized services

Advantages of Multi-Cloud

- High resilience (provider outages don’t stop system)
- Flexibility in service selection
- Better negotiation power with vendors
- Improved performance optimization

Challenges of Multi-Cloud

- Complex management
- Higher operational overhead
- Security consistency issues
- Skill requirements across platforms

Technologies Used in Multi-Cloud

- Kubernetes
- Terraform
- Service Mesh (Istio, Linkerd)
- CI/CD pipelines
- Centralized monitoring tools

Real-World Example

- Large E-commerce Platform
- AWS for backend services
- GCP for analytics
- Azure for identity management

Public vs Private vs Hybrid vs Multi-Cloud

Feature	Public Cloud	Private Cloud	Hybrid Cloud	Multi-Cloud
Definition	Shared cloud services over internet	Dedicated cloud for one organization	Combination of public + private cloud	Use of multiple cloud providers
Infrastructure Ownership	Cloud provider	Organization / Dedicated provider	Shared (public + private)	Multiple cloud providers
Tenancy	Multi-tenant	Single-tenant	Mixed	Mixed
Cost	Low (pay-as-you-go)	High	Medium	Medium–High
Scalability	Very high	Limited	High	Very high
Security	Moderate	Very high	High	High (complex)
Control	Low	Very high	High	Medium
Complexity	Low	Medium	High	High
Vendor Lock-in	High	Low	Medium	Low
Typical Users	Startups, students	Banks, government	Enterprises	Large enterprises
Use Cases	Web apps, SaaS	Sensitive data	Legacy + cloud apps	High availability systems
Examples	AWS, Azure, GCP	OpenStack, VMware	AWS + On-prem	AWS + Azure + GCP

Cloud Service Models

Cloud computing provides services at **three main levels**, based on how much control the user has.

- **IaaS** – Infrastructure as a Service
- **PaaS** – Platform as a Service
- **SaaS** – Software as a Service

IaaS

- **IaaS** is a cloud service model where **basic computing infrastructure** is provided over the internet.

It includes:

- Virtual machines
- Storage
- Networking

Users **manage the OS, applications, and data**, while the **cloud provider manages the hardware**.

What the Cloud Provider Manages

- Physical servers
- Storage devices
- Networking
- Data centers
- Virtualization layer

What the User Manages

- Operating system
- Runtime & middleware
- Applications
- Data
- Security configurations

Key Characteristics of IaaS

- On-demand resources
- High scalability
- Pay-as-you-go pricing
- Full control over infrastructure
- Suitable for customized setups

Examples of IaaS Providers

- AWS EC2
- Google Compute Engine
- Microsoft Azure Virtual Machines
- Oracle Cloud Infrastructure

Use Cases of IaaS

- Web application hosting
- Development and testing environments
- Disaster recovery
- Big data processing
- Running legacy applications

Advantages of IaaS

- No hardware investment
- Flexible and scalable
- Full control over servers
- Suitable for DevOps and CI/CD

Disadvantages of IaaS

- User responsible for OS and security
- Requires technical expertise
- More management compared to PaaS/SaaS

IaaS vs On-Premise

IaaS	On-Premise
Pay-as-you-go	High upfront cost
Highly scalable	Limited scalability
Fast provisioning	Slow setup
Provider managed hardware	Self managed hardware

SaaS

- **SaaS** is a cloud service model where **complete software applications** are delivered over the internet.
- Users **access the software via a web browser**, and **do not manage** servers, operating systems, or application updates.

What the Cloud Provider Manages

- Infrastructure (servers, storage, networking)
- Operating system
- Application software
- Security patches & updates
- Availability and scalability

What the User Manages

- User data
- Application configuration
- Access control (users & roles)

Key Characteristics of SaaS

- Ready-to-use applications
- No installation required
- Subscription-based or free
- Automatic updates
- Accessible from any device

Examples of SaaS Applications

- **Google Workspace (Gmail, Docs)**
- **Microsoft 365**
- **Salesforce**
- **Zoom**
- **Dropbox**

Use Cases of SaaS

- Email and communication
- Customer relationship management (CRM)
- Online collaboration
- E-learning platforms
- Accounting and finance software

Advantages of SaaS

- No infrastructure management
- Quick setup and use
- Lower upfront cost
- Easy scalability
- Always up-to-date

Disadvantages of SaaS

- Limited customization
- Data security and privacy concerns
- Vendor lock-in
- Internet dependency

SaaS vs Traditional Software

SaaS	Traditional Software
Cloud-based	Installed locally
Subscription model	One-time license
Auto updates	Manual updates
Accessible anywhere	Device dependent

PaaS

What Is PaaS?

- **PaaS** is a cloud service model that provides a **complete platform for application development, testing, and deployment**.
- Developers focus on **writing code**, while the cloud provider manages the **infrastructure and runtime environment**.

What the Cloud Provider Manages

- Servers, storage, networking
- Operating system
- Runtime environment
- Middleware
- Scaling & availability

What the User Manages

- Application code
- Application configuration
- Data

Key Characteristics of PaaS

- Developer-friendly
- Faster application development
- Built-in scalability
- Integrated development tools
- Supports CI/CD pipelines

Examples of PaaS Providers

- **Google App Engine**
- **AWS Elastic Beanstalk**
- **Microsoft Azure App Service**
- **Heroku**

Use Cases of PaaS

- Web application development
- API development
- Microservices applications
- Agile and DevOps workflows
- Student and startup projects

Advantages of PaaS

- No infrastructure management
- Faster development and deployment
- Reduced operational complexity
- Easy integration with cloud services

Disadvantages of PaaS

- Limited control over runtime
- Vendor lock-in
- Platform restrictions
- Less flexibility than IaaS

PaaS vs IaaS vs SaaS

Feature	IaaS	PaaS	SaaS
User manages	OS & apps	Apps & data	Only data
Flexibility	High	Medium	Low
Ease of use	Low	Medium	High
Target users	System admins	Developers	End users

Evolution of Cloud Computing

- Pre-Cloud Era
- Virtualization Era (2000–2005)
- Birth of Cloud Computing (2006)
- Growth Era (2010–2020)
- Modern Cloud (2020–Present)

https://www.geeksforgeeks.org/cloud-computing/evolution-of-cloud-computing/

Cloud Advantages - Benefits of Cloud Computing

- Trade fixed expense for variable expense
- Benefit from massive economies of scale
- Stop guessing capacity
- Increase speed and agility
- Stop spending money running and maintaining data centers
- Go global in minutes

Cloud Challenges

- Data Security and Privacy
- Cost Management
- Multi-Cloud Environments
- Performance Challenges
- Interoperability and Flexibility
- High Dependence on Network
- Lack of Knowledge and Expertise

Use Cases of Cloud

- Web & Mobile Application Hosting
- Data Storage & Backup
- Big Data Processing & Analytics
- Machine Learning & Artificial Intelligence
- Software Development & Testing
- Internet of Things (IoT)
- Disaster Recovery & Business Continuity
- Streaming & Content Delivery
- Enterprise Applications
- Education & E-Learning
- Gaming
- Government & Smart Cities

Modern Cloud

- Multi-cloud adoption
- AI-powered cloud services
- Edge computing
- Cloud security enhancements
- Fully managed services replacing in-house systems

Future of Cloud

- Quantum cloud computing
- Increased automation
- AI-driven infrastructure
- More edge computing
- Fully serverless systems
- Future is cloud-native + AI-powered

Q & A?